

SYEDA BUSHRA BANU

Full Stack Developer | Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 3+ years of professional experience, including 1.5 years building scalable, cloud-native web applications and RESTful APIs using Python, TypeScript, React, Next.js, and React Native. At Today's Dental Services, a multi-clinic dental group, built the Training & Certification, Bank Reconciliation and Provider Credentialing modules of its operations platform and maintained the group's clinic websites, using React/TypeScript, Python/FastAPI and PostgreSQL. Proficient in AWS (EC2, Lambda, S3), Firebase, MySQL, PostgreSQL, and CI/CD pipelines, with strong expertise in MVVM architecture and agile delivery. Passionate about writing clean, maintainable code and building impactful products across web and mobile platforms.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL

Frontend: React.js, Next.js (App Router), React Native (Expo, React Navigation), Astro, Vite, CRA, HTML5, CSS3, Responsive UI Design, Tailwind CSS, Bootstrap 5 / React Bootstrap, shadcn/ui, Zod, TanStack React Query, Jotai, react-hook-form, React Router, Recharts, jsPDF, html2canvas, react-signature-canvas, exceljs, react-calendar, SEO (JSON-LD, sitemaps, prerendering)

Backend & APIs: Python (Flask/FastAPI, httpx, psycopg 3, Alembic), Node.js, NestJS, Prisma ORM, REST APIs, API Integration, GraphQL & JSON-RPC integrations (Cherry, Odoo ERP), Open Dental REST API, Authorize.Net & Razorpay payment APIs (webhooks, refunds), Socket.IO / WebSockets, JWT auth (Passport), Google & Apple Sign-In, cron scheduling, rate limiting, Fernet / AES-256-GCM encryption, HMAC-signed URLs, Nodemailer, Expo push notifications, YouTube IFrame API, Amazon Chime SDK, Postman

Cloud & DevOps: AWS (EC2, Lambda, S3/boto3 presigned URLs, CloudFront, Amplify Hosting), Supabase (PostgreSQL, Storage), Firebase, Expo EAS, CI/CD Pipelines, GitHub Actions, Git, GitHub, Windows service deployment (WinSW), uvicorn

Databases: MySQL, PostgreSQL (Prisma ORM, Alembic migrations, psycopg 3, jsonb, row-level security), Supabase PostgreSQL, SQLite (node:sqlite), SQL, Firebase Realtime Database / Firestore, relational schema design & migrations

Architecture & Patterns: MVVM, MVC, Microservices, Serverless Architecture, Modular Monolith (NestJS), REST API Design, Real-time / Event-driven (WebSockets), Role-Based Access Control, Multi-clinic (multi-tenant) configuration, Static-site prerendering (JAMstack), Rule-based matching engines, Optimistic locking, Security hardening (SSRF guards, signed URLs, encryption at rest), LLM tool-calling agent loops

Tools & Platforms: SharePoint Framework (SPFx), SharePoint Designer, SharePoint Workflows, Jitsi Meet, Payment Gateway Integration, Playwright (browser automation, LLM-driven agent), pytest, Vitest, React Testing Library, Ruff, Axios, Prisma CLI, Alembic, tsup / esbuild, Excel & CSV data processing (exceljs, bank-statement parsing)

WORK EXPERIENCE

Full Stack Developer | Today's Dental Services

May 2025 - Present • Hyderabad, India

- Developed and maintained the group's patient-facing dental and medical practice websites (React/TypeScript, Astro, Vite), delivering API-backed appointment booking, patient-portal rescheduling, e-signed consent forms, a WebSocket AI chat widget and JSON-LD SEO.
- Built the React/TypeScript front end of a multi-clinic staff training and certification portal: role-assigned courses, onboarding tracks, anti-skip video tracking, sandboxed HTML courses, e-signed PDF certificates and certification-expiry monitoring.
- Extended the rule-based matching engine of the group's bank reconciliation module from 8 to 13 passes (remit groups, refund netting, check-to-EFT bridging), verifying that payments recorded in Open Dental reached the bank (Python/FastAPI, PostgreSQL).
- Integrated Odoo ERP bank feeds, Open Dental payments, Cherry (GraphQL) and Authorize.Net settlement APIs, Sunbit and CareCredit financing imports and bank-statement CSVs into reconciliation, plus a consolidated all-clinics summary grid (React/Next.js).
- Extended an on-premises Node.js/TypeScript Playwright agent to capture Cherry, Sunbit and CareCredit settlement reports through its LLM-driven navigation loop, keeping credentials out of model context and pushing normalized rows into reconciliation.
- Designed and built a provider credentialing system (Python/FastAPI, PostgreSQL, React/Next.js) replacing the spreadsheet that tracked dentists' insurance-payer enrollments, with a 9-status workflow, live NPI registry checks, spreadsheet import and expiry alerts.
- Backed the work with automated tests and typed contracts: 253 pytest and 432 Vitest cases on credentialing, 164 Vitest cases on the capture agent, Zod-validated API schemas and Alembic-managed PostgreSQL migrations.

Full Stack Development Intern | The Siasat Daily

March 2025 – May 2025 • Hyderabad, India

- Built dynamic web applications using HTML5, CSS3, JavaScript, and Python, delivering fully responsive front-end interfaces across desktop and mobile viewports.
- Developed server-side logic and database integrations using Python (Flask), handling form submissions and API requests for a news media platform with 1M+ monthly visitors.
- Integrated third-party REST APIs to power real-time content feeds and user authentication workflows.
- Gained hands-on exposure to full software development lifecycle (SDLC), including requirements gathering, implementation, testing, and deployment.

Data Curator Associate | JUKSHIO Technology Innovation Pvt Ltd

March 2023 – March 2025 • Hyderabad, India

- Curated and cleaned large-scale datasets to train AI/ML models, improving model accuracy by 15% through consistent annotation standards.
- Performed image annotation and dashboard labeling across 10,000+ data samples for computer vision model training pipelines.
- Conducted retail store testing and customer interaction analysis to validate AI-driven recommendation systems.
- Managed billing operations and produced feedback analysis reports that informed product improvement decisions.

PROJECTS

Training & Certification Module – Today's Dental Insights | React 18, TypeScript, Jotai, Axios, YouTube IFrame API, jsPDF, AWS S3 Link

- Built a training portal where admins publish video, document and sandboxed HTML courses, assign them by staff role, and onboard employees through role-based tracks and training manuals.
- Implemented segment-based anti-skip video tracking (YouTube IFrame API and HTML5 video) that auto-completes at 90% genuinely watched, with an engagement hook that pauses on idle or hidden tabs and syncs progress every 30 seconds.

- Shipped a 6-tab admin dashboard for enrollment, expiring-certification and billing analytics, a touch-enabled e-signature pad that exports PDF certificates (jsPDF, html2canvas), and a typed Axios layer over 72 REST endpoints.

Bank Reconciliation & Settlement Capture Module – Today's Dental Insights | Python (FastAPI), PostgreSQL, React/Next.js, TypeScript, Playwright

- Extended a Python/FastAPI reconciliation service (20 REST routes, 200+ pytest cases), growing its Open Dental-vs-Odoo matching engine from 8 to 13 passes with fee bands, remit groups, refund netting and check-to-EFT bridging.
- Integrated Cherry (GraphQL) and Authorize.Net settlement APIs and bank-statement CSV import (Bank of America, Chase, Odoo); built React/Next.js screens with a card-gateway switch, row-level match diagnostics and an all-clinics summary grid.
- Extended a Node.js/TypeScript on-prem Playwright agent, pairing its LLM navigation loop with deterministic credential handling to capture Cherry, Sunbit and CareCredit settlement reports for matching, covered by 164 Vitest tests.

Provider Credentialing Module – Today's Dental Insights | Python, FastAPI, PostgreSQL, Alembic, TypeScript, React, Next.js, exceljs, pytest/Vitest

- Designed a FastAPI/PostgreSQL backend (59 REST routes, 11 tables, 3 Alembic migrations) tracking dentist-to-insurance-payer enrollments per clinic with a 9-status workflow, change history, and admin-tier and dentist self-service access rules.
- Built a React/Next.js/TypeScript UI (10 screens, 9-step Add Provider wizard) and a 4-step browser-side xlsx import wizard (exceljs) with alias-normalised payer matching, driven by TypeScript rules and Zod schemas mirrored in Python.
- Integrated live CMS NPI Registry verification, Fernet encryption of SSNs and bank numbers, signed document downloads, and a 6-hourly push/email sweep alerting on expiring enrollments, licenses, malpractice policies and continuing-education certifications.

Healio – Healthcare Marketplace App | NestJS, Prisma, PostgreSQL (Supabase), React Native (Expo), Next.js, Socket.IO, Razorpay

- Built 17 of the 33 NestJS feature modules and 25 of the 57 Prisma models for a healthcare marketplace backend (Supabase PostgreSQL) connecting patients with doctors, hospitals, labs, pharmacies and insurers.
- Built the e-prescription, insurance-claims, home-care, field-agent and real-time chat modules and extended pharmacy ordering, including a Socket.IO messaging gateway and cron-scheduled appointment reminders.
- Integrated partner-owned Razorpay payment collection so each pharmacy, lab or hospital is paid into its own account: checkout, signature-verified webhooks, cash on delivery, scheduled refund retries and AES-256-GCM-encrypted gateway secrets.
- Hardened the API with HMAC-signed short-lived file links, per-IP rate limiting, magic-byte upload validation, SSRF and read-only SQL guards, weak-JWT-secret rejection and payment-capture verification.
- Built the React Native (Expo) patient and partner apps, authoring 57 of 65 and 37 of 69 screens, and solely authored a 52-page Next.js partner web portal with a GitHub Actions build workflow.

Clinic Websites – Today's Dental | React, TypeScript, Astro, Vite, Bootstrap, REST APIs, WebSockets, JSON-LD SEO, jsPDF [Link](#)

- Reworked React/TypeScript booking and patient-portal flows over the group's appointment API (patient lookup by name and birthdate, reschedule/cancel, document upload) and built consent e-signing that renders signed forms to PDF.
- Engineered a WebSocket AI-chat widget with automatic reconnect, heartbeat and iOS fixes, plus an SEO layer of schema.org JSON-LD data, per-route prerendered HTML and a build-time check enforcing title and description lengths.
- Delivered rollouts across the group's clinic sites (e.g. dentistinoregonoh.com, todaydentalexington.com) on three front-end stacks, including an insurance-plan comparison hub on 9 Astro sites and rockawayinternalmedicine.com built from scratch in Vite.

Dentipal – Dental Staffing Marketplace | React, TypeScript, AWS Lambda, DynamoDB, Amazon Cognito, API Gateway (WebSocket)

- Worked on a marketplace where dental clinics post jobs and dental professionals apply: React/TypeScript front end with an AI chat widget and WebSocket inbox on an AWS Lambda, DynamoDB and Cognito backend.
- Fixed clinic-side role-based access (Cognito groups and clinic membership), job-posting dashboard reads via DynamoDB secondary indexes, and professional-profile data bugs in the Find Professionals flow.

E-Commerce Platform with Payment Gateway | React.js, Python, MySQL, Razorpay API, AWS Lambda

- Developed a full-featured e-commerce web application with product catalog, shopping cart, and secure checkout using React.js and Python (Flask) REST APIs.
- Integrated Razorpay payment gateway, enabling end-to-end secure transactions with real-time payment status updates via webhooks.
- Deployed serverless order notification service on AWS Lambda, reducing infrastructure overhead by 40%.

Dreama – Pinterest Clone | HTML5, CSS3, JavaScript, Responsive Design

- Developed a responsive single-page web application replicating core Pinterest features: image upload, dynamic board creation, and content discovery grid.
- Achieved fully responsive layout across mobile, tablet, and desktop using CSS3 Flexbox/Grid with zero external UI frameworks.

EDUCATION

Master of Computer Applications (MCA) | Mesco Institute of Management & Computer Science, Hyderabad

June 2023 – May 2025

Bachelor of Commerce (B.Com) | Girraj Govt College, Telangana University, Nizamabad

May 2020 – May 2023

CERTIFICATIONS

- Soft Skills Training Certificate – Mahindra Pride Classroom by Nandi Foundation (Jan 2023 – Feb 2023)
- Youth Employment Programme (YEP) – Tata Consultancy Services (TCS) (Apr 2022 – Jun 2022)
- Post Diploma in Computer Applications – CSCE Computer Education, Nizamabad (Feb 2016 – Jan 2017)

ACHIEVEMENTS

- Completed NCC 'C' Certificate – Represented unit at EBSB National Camp; demonstrated leadership, team coordination, and civic responsibility (Feb 2022 – Jan 2023).
- Awarded NCC 'B' Certificate under Army Wing – Trained in leadership, physical endurance, and disaster management (Feb 2021 – Mar 2022).
- Silver Medalist, State Wrestling Championship – Demonstrated discipline, perseverance, and competitive excellence (May 2018).